

Lesson #5.00 2365 - Level 3 City and Guilds

Question one: Determine the reactance when an inductance of 40 mH is connected to a 230 V, 50 Hz supply? 1

Question two: Determine the reactance when a 120 μ F capacitor is connected to a 230 volt, 50 hertz supply. 3

Question one: Determine the reactance when an inductance of **40 mH** is connected to a **230 V**, 50 Hz supply?

1) Write the formula

$$X_L = 2 \times \pi \times \boxed{\text{F}} \times \boxed{\text{L}}$$

Hz H or mH

2) Swap in the values for the formula

$$2 \times 3.14 \times \boxed{50 \text{ Hz}} \times \boxed{0.040} = 12.566$$

-> 12.566 Ω (2 decimal places)
-> 12.57 Ω

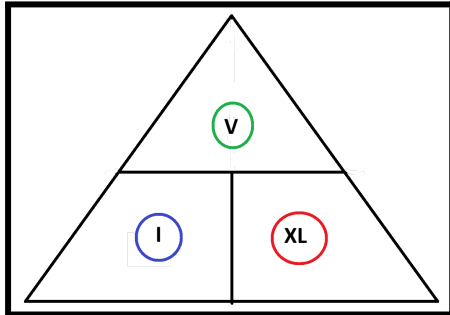
The answer to the question is **12.57 Ω** . For the question we want to find the resistance within the coils that is the definition of **X_L** .

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Question one: Additional steps to calculate current through the coils.

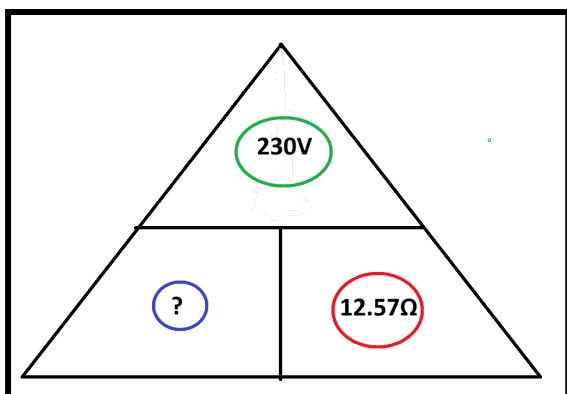
To calculate the current through the inductor/the coils. The steps are:

- 1) Write the detailed formula for Ohm's Law



- 2) Write the detailed formula for Ohm's Law. Swap in the values.

Remember, the voltage was provided in the question as **230 V** and the **XL** was calculated in the first part of the answer as **12.57Ω**.



Therefore, to calculate the Current/I we do **230V/12.57Ω**.

$$\frac{230 \text{ V}}{12.57} = 18.297 \text{ (2 D.P)}$$

18.30 amps

Question two: Determine the reactance when a **120 μ F capacitor** is connected to a **230 volt, 50 hertz** supply.

1) Display the formula

$$X_C = \frac{1}{2\pi fC}$$

$$2 \times \pi \times \boxed{\text{F}} \times \boxed{\text{C}}$$

Hz

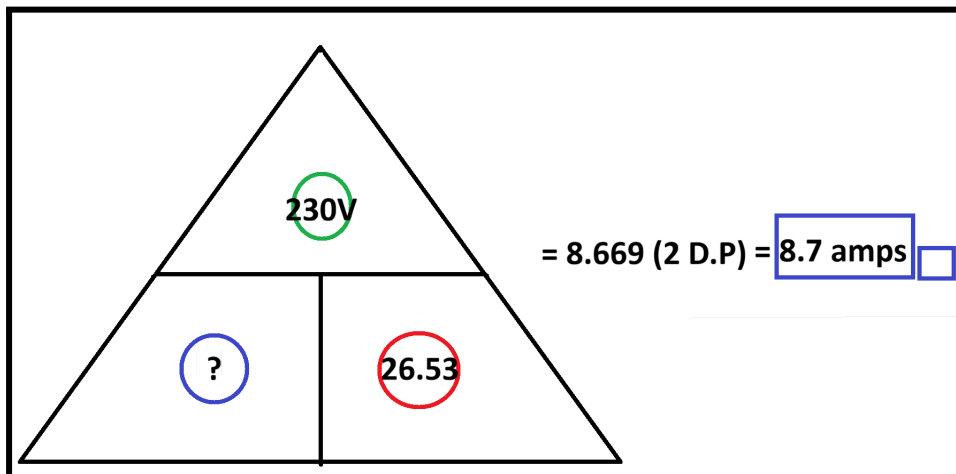
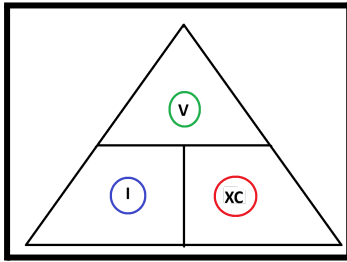
2) Swap in the values within the formula.

$$\frac{1}{2 \times \pi \times \boxed{50\text{Hz}} \times \boxed{120 \times 10^{-6}}} = 25.525 \text{ (2 D.P)}$$

$$\boxed{26.53}$$

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3) Additional steps to determine the current in amps.



- 1) The phases 120 degrees apart from one another.
- 2) L1, L2 and L3
- 3) Do not need the neutral in a balance system. There is no connection for the neutral.
- 4) In an unbalanced system you need a neutral. The neutral is there to take the current away.

Example one:

Star connection

A three-phase, four wire **star** connected transformer has a line voltage of **520 volts** and supplies a line current of **25 amps**. Calculate.

- a) The phase voltage of the transformer
- b) The phase current in each winding

- a) The phase voltage of the transformer

$$V_{\text{phase}} = \frac{V_{\text{line}}}{\sqrt{3}}$$

$$V_{\text{phase}} = \frac{520}{\sqrt{3}}$$

$$\sqrt{3} \approx 1.732$$

$$V_{\text{phase}} = \frac{V_L}{\sqrt{3}} = \frac{520}{1.732} \approx 300 \text{ V}$$

- b) The phase current in each winding

The answer for the question is relatively simple.
In a star relationship I_L equals I_P .

$$I_L = I_\phi$$

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= 25 amps.

Delta connection

Example two - A three-phase balanced delta connected resistive load is supplied from a transformer at a **line voltage of 400 volts** and draws a **line current of 13.86 amps**.
Calculate:

- a) A phase voltage
- b) The phase current
- c) The resistance of the load

A The Phase voltage

- a) In a phase voltage is equal to the line voltage.

$$V_{\phi} = V_L$$

- b) We know that the 400 Volts is the voltage of the line.

$$V_{\phi} = V_L = 400 \text{ V}$$

B The phase current

$$I_L = \sqrt{3} \times I_{\phi}$$

Rearrange the current to find I_{ϕ}

$$I_{\phi} = \frac{I_L}{\sqrt{3}} \Rightarrow \frac{13.86}{1.732} \approx 8 \text{ A}$$

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C The resistance of the load

The formula follows that of Ohms law.

$$R_{\phi} = \frac{V_{\phi}}{I_{\phi}}$$

Therefore:

$$R_{\phi} = \frac{400}{8}$$
$$R_{\phi} = 50 \Omega$$